

SYDNEY GRAMMAR SCHOOL



2015 Half-Yearly Examination

FORM V

MATHEMATICS EXTENSION 1

Monday 18th May 2015

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 80 Marks

- All questions may be attempted.

Section I – 8 Marks

- Questions 1–8 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 72 Marks

- Questions 9–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and Master on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your name, class and Master on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Nine.

5A: DS

5B: RCF

5C: SO

5D: DNW

5E: SJE

5F: REJ

5G: DWH

5H: KWM

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 125 boys

Examiner

SO

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

If $f(x) = \frac{3^x + 3^{-x}}{2}$, then $f(0)$ is:

- (A) 0
- (B) $\frac{1}{2}$
- (C) 1
- (D) 3

QUESTION TWO

Given $f(x) = \frac{1}{\sqrt[3]{x}}$, the correct expression for $f'(x)$ is:

- (A) $\frac{1}{\sqrt[3]{x^4}}$
- (B) $-\frac{1}{\sqrt[3]{x^4}}$
- (C) $\frac{4}{3\sqrt[3]{x^4}}$
- (D) $-\frac{1}{3\sqrt[3]{x^4}}$

QUESTION THREE

Consider the curve $f(x) = x^3 - ax$. The x -coordinates of the points where the tangent to the curve is horizontal are:

- (A) 0 and a
- (B) ± 1
- (C) $\pm \sqrt{\frac{a}{3}}$
- (D) $3 - a$ and $3 + a$

Examination continues next page ...

QUESTION FOUR

Which of the following is NOT an odd function?

(A) $f(x) = x^3$

(B) $f(x) = \frac{1}{x}$

(C) $f(x) = \sin x$

(D) $f(x) = \cos x$

QUESTION FIVE

Which statement is TRUE?

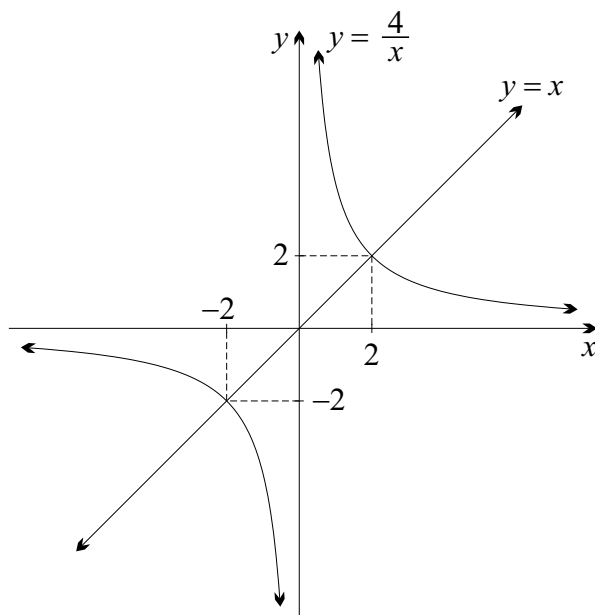
(A) $\sin(360^\circ - A) = -\sin A$

(B) $\cos(90^\circ - A) = \operatorname{cosec} A$

(C) $\cot A = \frac{\sin A}{\cos A}$

(D) $\operatorname{cosec}(180^\circ + A) = \frac{1}{\sin A}$

QUESTION SIX



Solve the inequation $\frac{4}{x} < x$ using the graph above:

- (A) $x < -2$ or $0 < x < 2$
- (B) $-2 < x < 0$ or $x > 2$
- (C) $-2 < x < 2$
- (D) $x < -2$ or $x > 2$

QUESTION SEVEN

The expression $\frac{\frac{1}{x^2} - \frac{1}{y^2}}{x + y}$ simplifies to:

- (A) $\frac{x - y}{x^2 y^2}$
- (B) $\frac{y - x}{x^2 y^2}$
- (C) $\frac{x - y}{xy}$
- (D) $\frac{-y - x}{xy}$

QUESTION EIGHT

The interval AB has endpoints $A(-1, 4)$ and $B(x, y)$. The point $P(14, -6)$ divides the interval AB externally in the ratio $5 : 3$. Find the value of x .

- | | |
|-----------|---------------------|
| (A) 5 | (B) $\frac{31}{5}$ |
| (C) 23 | (D) $\frac{109}{5}$ |

————— End of Section I —————

Examination continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION NINE (12 marks) Use a separate writing booklet.	Marks
(a) Solve $\frac{1}{m} + 6 = 1 - \frac{4}{m}$.	1
(b) Evaluate $\sum_{k=2}^4 3^k$.	1
(c) Find the limiting sum of the series $1 + \frac{1}{3} + \frac{1}{9} + \dots$	1
(d) Differentiate:	
(i) $2x^3 + 5$	1
(ii) $\frac{3}{x}$	1
(iii) $(x + 1)(x - 2)$	1
(e) A line has angle of inclination 120° . Find its gradient.	1
(f) The line $x - y - 17 = 0$ is a tangent to a circle with centre $(2, -3)$.	
(i) Find the perpendicular distance from the centre of the circle to the tangent.	2
(ii) Hence write down the equation of the circle.	1
(g) Find the equation of the tangent to the curve $y = x^2 - 8x$ at the point $P(-1, 9)$.	2

QUESTION TEN (12 marks) Use a separate writing booklet. **Marks**

- (a) An arithmetic sequence has second term 11 and eighth term -40 . Find the common difference and the first term. **3**
- (b) Solve the inequation $|2x + 3| > 1$. **2**
- (c) Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$:
- (i) $\sqrt{3} \sin \theta = \cos \theta$ **2**
- (ii) $\sec^2 \theta + \tan \theta = 1$ **3**
- (d) In triangle XYZ , $\angle XYZ = 30^\circ$, $XZ = a\sqrt{2}$ and $YZ = 2a$. Find the two possible values of $\angle ZXY$. **2**

QUESTION ELEVEN (12 marks) Use a separate writing booklet. **Marks**

- (a) Differentiate:
- (i) $\frac{2x + x^3}{x}$ **2**
- (ii) $(8 + 5x)^3$ **2**
- (iii) $\frac{x^2 + 3}{1 - x}$ **2**
- (b) Consider the sequence for which $T_n = 3n + 3^n$.
- (i) Find the first three terms of this sequence. **1**
- (ii) Find the sum of the first eight terms. **3**
- (c) Prove the identity $\cot \theta \cos \theta + \sin \theta = \operatorname{cosec} \theta$. **2**

QUESTION TWELVE (12 marks) Use a separate writing booklet.**Marks**(a) Sketch the graph of $y = -|x|$. 1(b) Consider the function $f(x) = 3x^2 - 4x$.(i) Simplify the expression $f(x + h) - f(x)$. 2(ii) Hence use the formula 1

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

to differentiate $f(x) = 3x^2 - 4x$ from first principles.(c) Consider the graphs of $(x - 4)^2 + y^2 = 9$ and $y = x - 7$.(i) Determine the points of intersection. 2(ii) Shade the region where both $(x - 4)^2 + y^2 \leq 9$ and $y > x - 7$ are satisfied. 3(d) The lines $\ell_1 : 4x - y - 5 = 0$ and $\ell_2 : 3x + 4y + 1 = 0$ intersect at the point M .(i) Write down the general equation of a line through M , and show that it can be written in the form 1

$$(3k + 4)x + (4k - 1)y + (k - 5) = 0$$

for some constant k .(ii) Hence find the line through M and the point $P(6, 0)$. 2

QUESTION THIRTEEN (12 marks) Use a separate writing booklet.**Marks**

(a) Solve the inequation $\frac{x-1}{x+2} \geq 2$.

3

(b) Suppose that $\tan \alpha = \frac{a^2 - b^2}{2ab}$, where α is acute, and both a and b are positive.
Find an expression for $\cos \alpha$.

2

(c) Consider the curve $y = \frac{x^2 - 1}{(x-3)(x+2)}$.

(i) Write down the equation(s) of any vertical asymptote(s).

1

(ii) Find the equation of the horizontal asymptote.

1(iii) Find any intercepts with the x and y axes.**2**

(iv) Copy and complete the following table:

1

x	-4	$-\frac{3}{2}$	2	4
y				

(v) Hence draw a neat sketch of the curve.

2**QUESTION FOURTEEN** (12 marks) Use a separate writing booklet.**Marks**

(a) Show that for any non-zero polynomial $Q(x)$, the curve $y = (x-a)^2 Q(x)$ has a horizontal tangent at $x = a$.

2

(b) Consider the function $y = 2|x+1| + |x-5|$.

(i) Sketch the graph of the function by considering separately
 $x \geq 5$, $x < -1$ and $-1 \leq x < 5$.**3**(ii) Hence, or otherwise, solve the inequation $2|x+1| + |x-5| < 15$.**1**

(c) Consider the function $f(x) = \frac{2}{x^2 + 4x + 8}$.

(i) Find the domain.

1

(ii) Find the range.

2

(d) If the sum of the first p terms of an arithmetic series is equal to zero, show that the sum of the next q terms is $\frac{aq(p+q)}{1-p}$, where a is the first term.

3

 End of Section II

END OF EXAMINATION

B L A N K P A G E

NAME:

CLASS: MASTER:

SYDNEY GRAMMAR SCHOOL



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

2015 FIFTH FORM 3 UNIT HALF-YEARLY

SOLUTIONS

1. C
2. D
3. C
4. D
5. A
6. B
7. B
8. A

QUESTION 9

$$(a) \frac{1}{m} + 6 = 1 - \frac{4}{m}$$

$$\frac{5}{m} = -5$$

$$m = -1 \quad \checkmark$$

$$(b) \sum_{k=2}^4 3^k = 3^2 + 3^3 + 3^4$$
$$= 9 + 27 + 81$$
$$= 117 \quad \checkmark$$

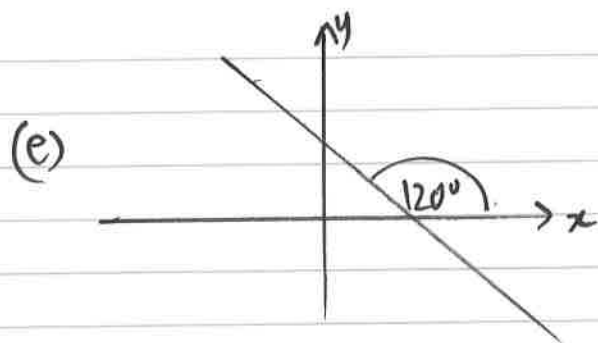
$$(c) S_{\infty} = \frac{1}{1 - \frac{1}{3}}$$

$$= \frac{3}{2} \quad \checkmark$$

$$(d)(i) \frac{d}{dx}(2x^3 + 5) = 6x^2 \quad \checkmark$$

$$(ii) \frac{d}{dx}\left(\frac{3}{x}\right) = -\frac{3}{x^2} \quad \checkmark$$

$$(iii) \frac{d}{dx}(x+1)(x-2) = \frac{d}{dx}(x^2 - x - 2)$$
$$= 2x - 1 \quad \checkmark$$



$$m = \tan 120^\circ$$

$$= -\sqrt{3}$$

$\therefore$ the gradient is $-\sqrt{3}$ ✓

(f) (i) perpendicular distance = $\frac{|1(2) - 1(-3) - 17|}{\sqrt{1^2 + (-1)^2}}$ ✓

$$= \frac{|2 + 3 - 17|}{\sqrt{2}}$$

$$= \frac{12}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$
 ✓

$$= 6\sqrt{2} \text{ units}$$

(ii) $(x-2)^2 + (y+3)^2 = (6\sqrt{2})^2$

$$(x-2)^2 + (y+3)^2 = 72$$
 ✓

(g) $y = x^2 - 8x$

$$\frac{dy}{dx} = 2x - 8$$

at $x = -1$, $\frac{dy}{dx} = 2(-1) - 8$

$$= -10$$

equation of tangent is

$$y - 9 = -10(x + 1)$$

$$y - 9 = -10x - 10$$

$$10x + y + 1 = 0$$
 ✓

(or $y = -10x - 1$)

QUESTION 10

(a) $a + d = 4$ (1) ✓
 $a + 7d = -40$ (2)

(2) - (1) $6d = -51$ ✓
 $d = -8\frac{1}{2}$

sub back into (1): $a = 11 + 8\frac{1}{2}$ ✓
 $= 19\frac{1}{2}$

$\therefore a = 19\frac{1}{2}$
 $d = -8\frac{1}{2}$

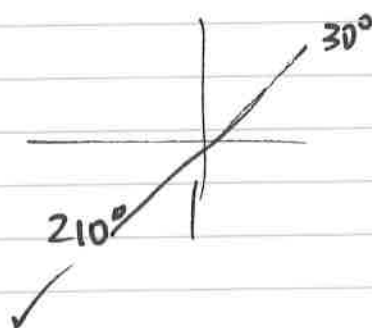
(b) $|2x + 3| > 1$

$2x + 3 > 1$
 $2x > -2$
 $x > -1$ ✓

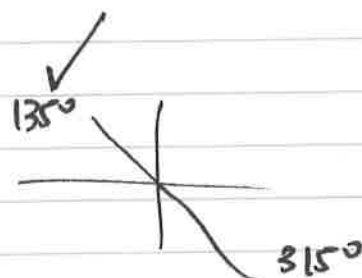
or $2x + 3 < -1$
 $2x < -4$
 $x < -2$ ✓

so $x < -2$ or $x > -1$.

(c) (i) $\sqrt{3} \sin \theta = \cos \theta$
 $\frac{\sin \theta}{\cos \theta} = \frac{1}{\sqrt{3}}$
 $\tan \theta = \frac{1}{\sqrt{3}}$ ✓
 $\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$
 $= 30^\circ, 210^\circ$

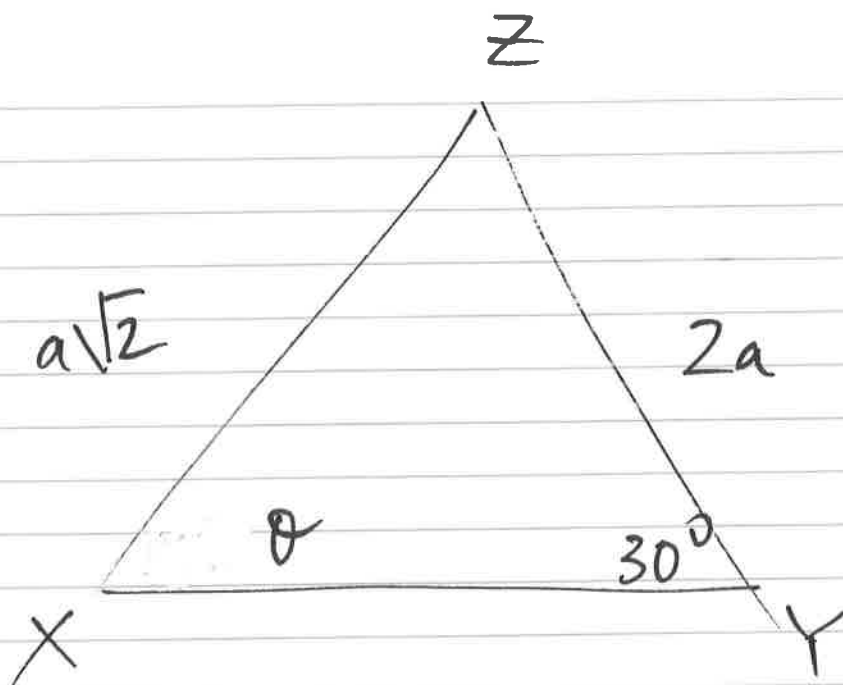


(ii) $\sec^2 \theta + \tan \theta = 1$
 $1 + \tan^2 \theta + \tan \theta = 1$
 $\tan^2 \theta + \tan \theta = 0$
 $\tan \theta (\tan \theta + 1) = 0$
 $\tan \theta = 0, \tan \theta = -1$
 $\theta = 0^\circ, 180^\circ, 360^\circ$
 acute $\theta = 45^\circ$



$\therefore \theta = 0^\circ, 135^\circ, 180^\circ, 315^\circ, 360^\circ$ ✓

(d)



$$\frac{\sin \theta}{2a} = \frac{\sin 30^\circ}{a\sqrt{2}} \quad \checkmark$$

$$\sin \theta = \frac{2a}{2a\sqrt{2}}$$

$$\sin \theta = \frac{1}{\sqrt{2}}$$

$$\theta = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$= 45^\circ$$

θ could also $= 135^\circ$ ($135 + 30 = 165 < 180$)

$$\therefore \theta = 45^\circ, 135^\circ \quad \checkmark$$

QUESTION 11

$$(a)(i) \frac{d}{dx} \left(\frac{2x+x^3}{x} \right) = 2x \quad \checkmark \checkmark$$

$$(ii) \frac{d}{dx} \left((8+5x)^3 \right) = 3(8+5x)^2 \times 5 \quad \checkmark$$
$$= 15(8+5x)^2 \quad \checkmark$$

$$(iii) \frac{d}{dx} \left(\frac{x^2+3}{1-x} \right) = \frac{2x(1-x) + x^2+3}{(1-x)^2} \quad \checkmark$$
$$= \frac{2x - 2x^2 + x^2 + 3}{(1-x)^2}$$
$$= \frac{-x^2 + 2x + 3}{(1-x)^2} \quad \checkmark$$

$$(b)(i) T_n = 3n + 3^n$$

$$\left. \begin{array}{l} T_1 = 3 + 3 \\ = 6 \\ T_2 = 6 + 3^2 \\ = 15 \\ T_3 = 9 + 3^3 \\ = 36 \end{array} \right\} \quad \checkmark$$

(ii) $3n$ is an AP with $a = 3$ and $l = 24, n = 8$
so let its sum be S_a .

$$S_a = \frac{8}{2} (3 + 24)$$
$$= 108 \quad \checkmark$$

3^n is a GP with $a = 3, r = 3, n = 8$; let its sum be S_G .

$$S_G = \frac{3(3^8 - 1)}{3 - 1}$$

$$= 9840 \quad \checkmark$$

$$\text{So } S_g = S_a + S_g$$

$$= 108 + 9840$$

$$= 9948 \quad \checkmark$$

$$(c) \text{ Prove } \cot \theta \cos \theta + \sin \theta = \operatorname{cosec} \theta$$

$$\text{LHS} = \cot \theta \cos \theta + \sin \theta$$

$$= \frac{\cos \theta}{\sin \theta} \cdot \cos \theta + \sin \theta$$

$$= \frac{\cos^2 \theta}{\sin \theta} + \sin \theta \quad \checkmark$$

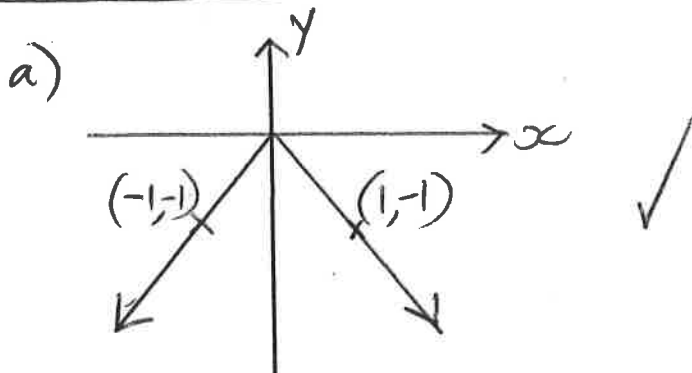
$$= \frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta}$$

$$= \frac{1}{\sin \theta}$$

$$= \operatorname{cosec} \theta \quad \checkmark$$

$$= \text{RHS}$$

QUESTION 12



b) $f(x) = 3x^2 - 4x$

(i) $f(x+h) - f(x) = 3(x+h)^2 - 4(x+h) - [3x^2 - 4x]$ ✓
 $= 3x^2 + 6xh + 3h^2 - 4x - 4h - 3x^2 + 4x$
 $= h(6x - 4 + 3h)$ ✓

(ii) $f'(x) = \lim_{h \rightarrow 0} (6x - 4 + 3h)$
 $= 6x - 4$ ✓

c) $(x-4)^2 + y^2 = 9$ $y = x - 7$ ✓

(i) $\therefore (x-4)^2 + (x-7)^2 = 9$ ✓
 $x^2 - 8x + 16 + x^2 - 14x + 49 = 9$

$$2x^2 - 22x + 56 = 0$$

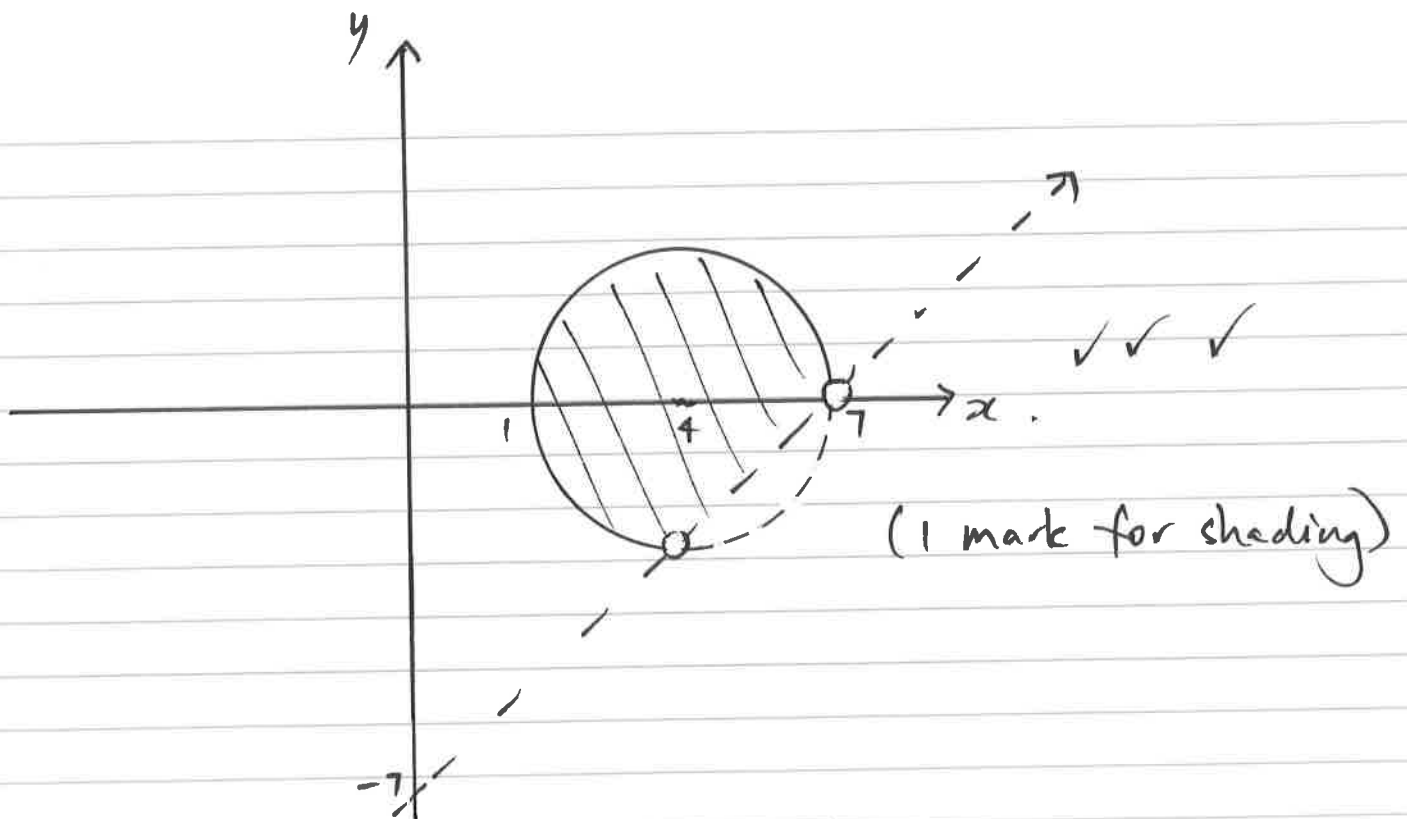
$$x^2 - 11x + 28 = 0$$

$$(x-7)(x-4) = 0$$

$$\therefore x = 7 \text{ or } 4$$

$$y = 0 \text{ or } -3$$

so $(7, 0)$ and $(4, -3)$ ✓



d(i) $4x - y - 5 + k(3x + 4y + 1) = 0$
 $4x - y - 5 + 3kx + 4ky + k = 0$
 $(3k + 4)x + (4k - 1)y + (k - 5) = 0$ ✓

(ii) P is a line to
 $(3k + 4) \cdot 6 + (4k - 1) \cdot 0 + k - 5 = 0$
 $18k + 24 + k - 5 = 0$
 $19k + 19 = 0$
 $k = -1$ ✓

So the equation of the line is

$(3(-1) + 4)x + (4(-1) - 1)y + (-1 - 5) = 0$
 $x - 5y - 6 = 0$ ✓

QUESTION 13

(a) $\frac{x-1}{x+2} \geq 2 \quad (x \neq -2)$

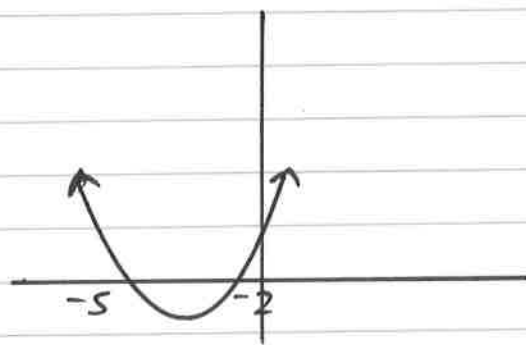
$$(x-1)(x+2) \geq 2(x+2)^2 \quad \checkmark$$

$$2(x+2)^2 - (x+2)(x-1) \leq 0$$

$$(x+2)[2x+4 - x+1] \leq 0$$

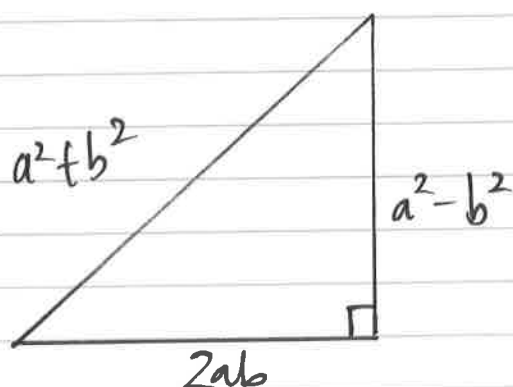
$$(x+2)(x+5) \leq 0 \quad \checkmark$$

$$-5 \leq x < -2 \quad \checkmark$$



(b) $a, b > 0$

$$\tan \alpha = \frac{a^2 - b^2}{2ab}$$



By Pythagoras' theorem:

$$= (a^2 - b^2)^2 + (2ab)^2$$

$$= a^4 - 2a^2b^2 + b^4 + 4a^2b^2$$

$$= a^4 + 2a^2b^2 + b^4$$

$$= (a^2 + b^2)^2$$

$$\text{hypotenuse} = a^2 + b^2 \quad \checkmark$$

$$\cos \alpha = \frac{2ab}{a^2 + b^2} \quad \checkmark$$

$$(i) y = \frac{x^2 - 1}{(x-3)(x+2)}$$

$$(i) x=3 \text{ and } x=-2 \quad \checkmark$$

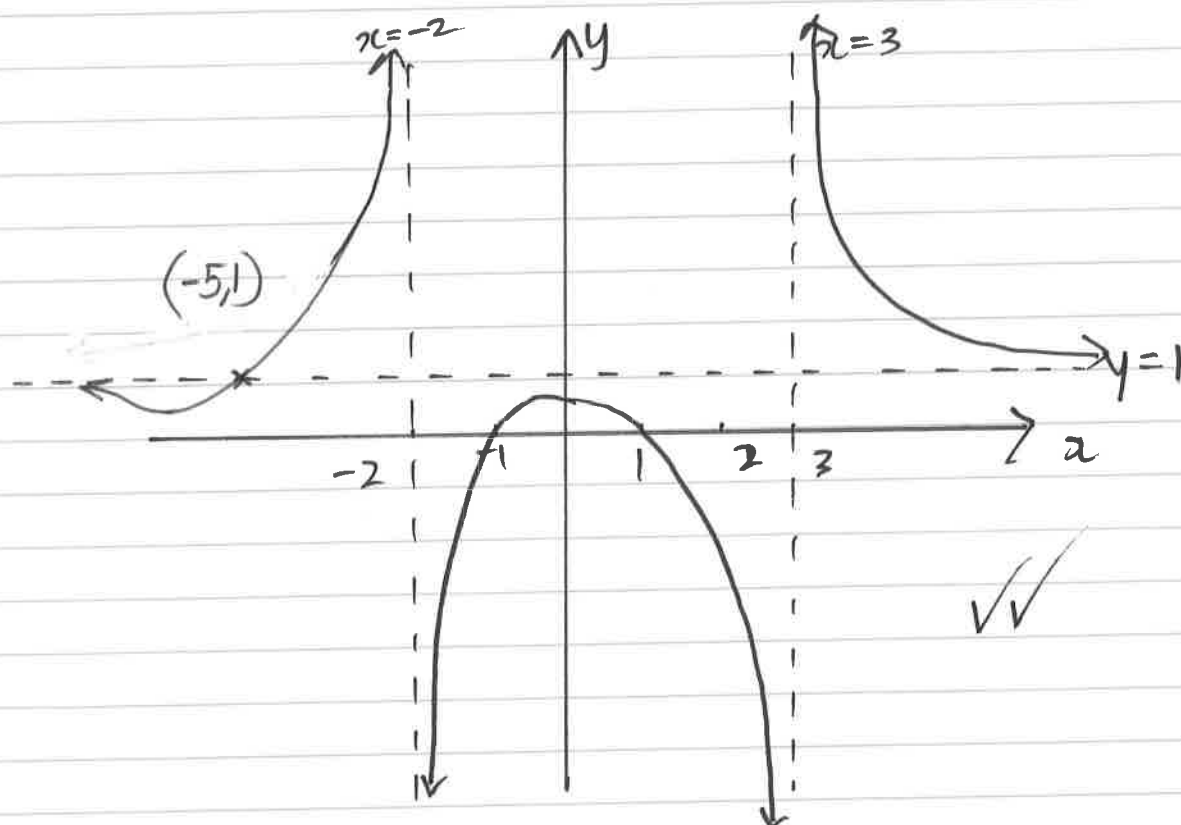
$$(ii) y = \frac{\frac{x^2}{x^2} - \frac{1}{x^2}}{\frac{x^2}{x^2} - \frac{1}{x^2} - \frac{6}{x^2}} = \frac{1 - \frac{1}{x^2}}{1 - \frac{1}{x} - \frac{6}{x^2}}$$

Horizontal asymptote $y=1 \quad \checkmark$

$$(iii) y\text{-intercept: When } x=0, y = \frac{-1}{-3 \times 2} = \frac{1}{6} \quad (0, \frac{1}{6}) \quad \checkmark$$

$$x\text{-intercept: when } y=0, x^2 - 1 = 0 \quad x=1 \text{ or } -1 \quad (1, 0) \text{ and } (-1, 0) \quad \checkmark$$

$$(iv) \begin{array}{c|c|c|c|c} x & -4 & -\frac{3}{2} & 2 & 4 \\ \hline y & \frac{15}{4} & -\frac{5}{4} & -\frac{3}{4} & \frac{15}{6} \end{array} \quad \checkmark$$



(NB: Left hand branch actually crosses horizontal asymptote at $(-5, 1)$ and approaches from below - No penalty for students not recognising this)

QUESTION 14

(a) $f(x) = (x-a)^2 \cdot Q(x)$

$f'(x) = 2(x-a) \cdot Q(x) + Q'(x)(x-a)^2$ ✓

at $x=a$, $f'(a) = 2(a-a)Q(x) + Q'(x)(a-a)^2$
 $= 2 \times 0 + 0$
 $= 0$

So the polynomial has a horizontal tangent at $x=a$ ✓

(b)(i) when $x \geq 5$, $y = 2x+2+x-5$
 $= 3x-3$

when $-1 \leq x < 5$, $y = 2x+2-x+5$
 $= x+7$

when $x < -1$, $y = -2x-2-x+5$
 $= -3x+3$

✓ (branches)

Points of Intersection

$3x-3 = x+7$

$2x = 10$

$x = 5$

when $x=5$, $y=12$

$(5,12)$

$-3x+3 = x+7$

$-4 = 4x$

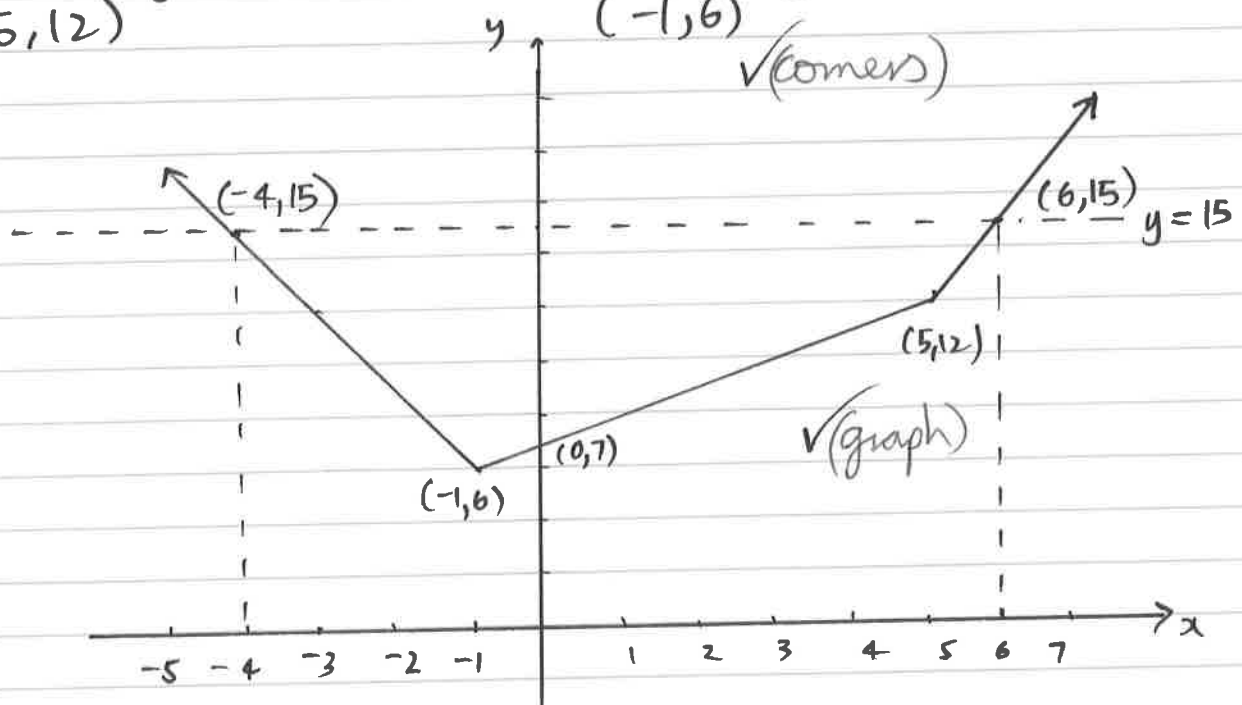
$x = -1$

when $x=-1$, $y=6$

$(-1,6)$

y-intercept
 $(0,7)$

✓ (corners)



(iv) $-4 < x < 6$ ✓

$$(c) \ f(x) = \frac{2}{(x^2+4x+4)+4}$$

$$= \frac{2}{(x+2)^2 + 4}$$

(i) Domain : all real x ✓

(ii) when $x = 0$, $y = \frac{1}{2}$

Range : $0 < y \leq \frac{1}{2}$ ✓✓

$$(d) \ \underbrace{a + a+d + a+2d + \dots + a + (p-1)d}_{p \text{ terms}}$$

$$S_p = \frac{p}{2}(2a + (p-1)d) = 0 \quad (*)$$

$$p \neq 0, \text{ so } 2a + (p-1)d = 0 \quad \checkmark$$

$$d = -\frac{2a}{p-1}$$

$$= \frac{2a}{1-p}$$

$$S_{p+q} = \frac{p+q}{2} [2a + (p+q-1)d]$$

$$= \frac{p+q}{2} [2a + (p-1)d + qd]$$

$$= \frac{p+q}{2} \cdot qd \quad (\text{using } (*)) \quad \checkmark$$

$$\text{Also from } (*) \ d = \frac{2a}{1-p}.$$

$$\begin{aligned}
 \text{So } S_{p+q} &= \frac{p+q}{2} \cdot q \cdot \frac{2a}{1-p} \\
 &= \frac{2aq(p+q)}{2(1-p)} \\
 &= \frac{aq(p+q)}{1-p}
 \end{aligned}$$

}

Since $S_p = 0$, S_{p+q} is the sum of the next q terms.